

Appliance Repair Training

Washing Machine & Refrigerator Repair and Service

10-DAY PROFESSIONAL PROGRAM



Made with GAMMA

DAY 1

Basic Electrical & Safety

1

Electrical Safety Rules

Foundational safety protocols every technician must know before handling any appliance. Safe work habits that prevent injury and equipment damage on the job.

2

AC & DC Basics

Understanding alternating current and direct current — how they behave, where they appear in household appliances, and why the distinction matters for diagnosis and repair.

3

Voltage, Current, Resistance & Multimeter Introduction

Core concepts of Ohm's Law applied to real-world appliance circuits. Introduction to the digital multimeter as the technician's most essential diagnostic tool.

Tools Handling & Testing Instruments

Digital Multimeter & Clamp Meter

Hands-on handling of the **digital multimeter** for measuring voltage, resistance, and continuity. Clamp meter techniques for measuring current draw without breaking the circuit — essential for motor and compressor diagnosis.

Screwdriver Set & Wiring Tester

Correct usage of **insulated screwdriver sets** for safe appliance disassembly. Wiring tester application for identifying live wires, broken circuits, and incorrect connections in washing machine and refrigerator wiring harnesses.

Electrical Safety Equipment

Personal protective equipment (PPE) selection and correct usage. Insulated gloves, safety goggles, and anti-static precautions that form the non-negotiable foundation of professional appliance service work.

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Introduction to Washing Machines – Day 3

Trainees gain a thorough understanding of both major washing machine categories, learn to identify every key component by sight, and begin reading real wiring diagrams used in the field.

2

Machine Types

Top Load & Front Load — each with unique drum mechanics, control systems, and common failure patterns

10+

Components Identified

Motor, drum, door seal, water inlet valve, control board, drain pump, timer, lid switch, and more

Day 3

Wiring Diagrams

Reading and interpreting basic wiring diagrams to trace circuits and understand how components interact

"Understanding the machine's anatomy is the difference between guessing and diagnosing. Every repair starts with knowing the components."

— Day 3 Training Objective

DAY 4 & 5

Washing Machine Testing, Diagnosis & Repair



Components Testing – Day 4

Systematic testing of the **water inlet valve, drain pump, capacitor, and motor** using a multimeter. Technicians learn expected readings for healthy components and how to identify a failed part with confidence.



Fault Diagnosis – Spin, Fill & Drainage

Spin problems: diagnosing motor faults, capacitor failure, lid switch issues.
Water fill problems: inlet valve blockage, water pressure faults, pressure switch testing. **Drainage problems:** pump obstruction, hose kinking, pump motor failure.



Complete Machine Servicing – Day 5

End-to-end practical servicing of a full washing machine unit. From disassembly to component testing, fault isolation, replacement, reassembly, and final functional verification — the complete professional workflow.

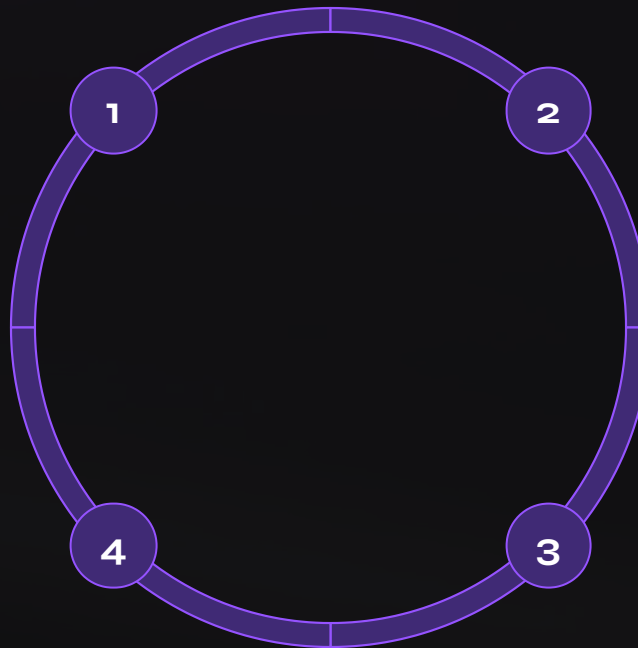
Refrigerator Basics

Refrigeration Cycle

How refrigerant circulates through evaporator, condenser, compressor, and expansion valve to transfer heat and create cooling.

Cooling System Overview

Understanding the full cooling system architecture — single-door, double-door, and frost-free designs and how their layouts affect service procedures.



Compressor Working Principle

The compressor as the heart of the system — how it pressurizes refrigerant and what electrical and mechanical signals indicate health or failure.

Thermostat Operation

How the thermostat senses temperature and switches the compressor on and off to maintain the set cooling level inside the cabinet.

Refrigerator Components & Testing – Day 7

Precision Testing Builds Confident Technicians



Day 7 puts testing instruments in trainees' hands for every critical refrigerator component. Each test builds a clear picture of what a **healthy reading looks like** versus a faulty one.

1

Compressor Testing

Resistance checks across compressor terminals. Identifying open windings, shorted windings, and compressor-to-ground faults using a multimeter.

2

Relay & Overload Protector Testing

Start relay shake test and resistance check. Overload protector continuity testing — understanding when each component must be replaced to restore starting function.

3

Fan Motor Testing

Evaporator and condenser fan motor winding resistance checks. Blade obstruction inspection. Understanding how fan failure causes compressor overheating and cooling loss.

Refrigerator Fault Diagnosis, Repair & Complete Service

Two days of intensive practical work covering the most common refrigerator faults technicians encounter in the field, progressing to a full end-to-end professional service procedure.



Cooling Problems

Diagnosing insufficient cooling — compressor not starting, dirty condenser coils, refrigerant loss indicators, and thermostat miscalibration. Practical component replacement procedures.



Water Leakage Problems

Identifying internal and external water leaks. Blocked drain tube clearing, door gasket seal inspection, and drip tray overflow causes with step-by-step remedies.



Ice Formation Issues

Diagnosing excessive ice or frost build-up in frost-free models. Defrost heater testing, defrost timer or board faults, and door seal failures causing humidity ingress.



Complete Service — Day 9

Full refrigerator maintenance workflow: cleaning, component inspection, performance testing, customer complaint analysis, and practical troubleshooting of real-world units.

Business Setup & Final Assessment

1 Service Business Setup

How to register a repair service, set up a home workshop or mobile service unit, and source spare parts reliably. Building a professional brand from day one.

2 Customer Handling & Service Charge Calculation

Professional communication skills for managing customer expectations, explaining faults clearly, and quoting jobs confidently. Labour + parts pricing methodology for sustainable margins.

3 Final Practical Assessment & Certificate Distribution

Trainees demonstrate end-to-end diagnosis and repair on a real appliance unit under assessment conditions. Successful graduates receive their **professional training certificate** to begin their career.

10-Day Training at a Glance

A structured theory-plus-practical journey from foundational electrical knowledge through to professional appliance service competency and business readiness.

